Mechanical design review

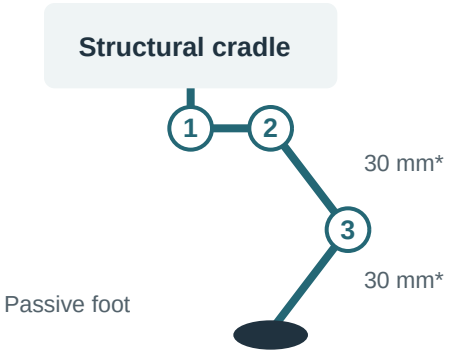
A compact four-legged robot in a retro Macintosh-shaped clock body

Review goal: Find a practical joint arrangement that preserves Mack's short-leg appearance. Resolve leg mechanics before docking controls. This is an early concept; no fitted complete leg or physical prototype exists yet.

DONOR BODY / CATALOG [1]	DONOR CLOCK / CATALOG [1]	PROPOSED LEG ACTUATION
80 x 91 x 112 mm	**229 g / 8.08 oz**	**3 axes x 4 legs = 12**

Working W x D x H order is provisional. Actual interior, retained shell mass and completed robot mass are unknown.

PROPOSED JOINT CHAIN



Schematic. *Proposed axis-to-axis link lengths.

1 Hip spread: moves the leg sideways on a fore/aft axis.

2 Hip swing: moves the upper leg forward/back on a transverse axis.

3 Knee: bends the lower leg; parallel to the hip-swing axis.

Load path: foot, links, supported shafts and bearings, then cradle. The cosmetic shell and motor horn alone must not carry leg loads.

Retain a compliant foot with limited passive tilt. Allow space for stops, cable loops, bearing retention and service access.

MAIN DECISION: WHERE DOES THE KNEE MOTOR GO?

A / Motor at the knee

First prototype baseline. Short drive path and fewer transmission parts; adds moving mass and bulk to the upper leg.

B / Motor beside the hip

Belt or linkage drives the knee. Moves motor mass inward; adds routing, supports, tensioning and possible motion coupling.

Scale check [2]: A ROBOTIS XL330-M288-T case is 20 x 34 x 26 mm and 18 g. Twelve bare actuators total 216 g. Its 34 mm case dimension is already larger than our 30 mm link pitch. This is a size benchmark, not a selected or load-qualified motor; published stall torque is not continuous capacity.

FEEDBACK REQUESTED

1. Which knee-drive layout best fits this scale without losing the short-leg appearance?

2. What bearing, shaft and cradle arrangement would you use to carry side loads and impacts?

3. What assembled mass, joint torque and duty-cycle assumptions should govern motor selection?

4. What would you change before building one leg? Self-righting reach is still unproven.

Next proof: Verify the underside geometry and retained mass; fit real motor, bearing and cable envelopes into one leg; then test that leg in a restrained fixture under representative loads.

Sources checked 22 Oct 2026. Proposals are Mack design-study assumptions.

[1] RayCue WB-8: raycue.com/products/maclock
[2] ROBOTIS: emanual.robotis.com/docs/en/dxl/x/xl330-m288/